

1. Data Understanding

1.1 Load the dataset

The health_data.csv was loaded using Pandas read_csv

1.1.1 Check the first few entries

Checked first five rows using df.head()

1.1.2 Remove columns which are irrelevant

Irrelevant columns like ['Unnamed: 0', 'id'] were dropped

1.1.3 Inspect the shape of the dataset

Dataset has 70,000 patients and 12 columns: Features: 11 and Target: 1 (cardio)

1.1.4 Inspect the different columns in the dataset

1. There are no missing values (all 12 columns show 70000 non-null) => no imputation needed
2. ['age', 'gender', 'height', 'weight', 'ap_hi', 'ap_lo'] are float64 meaning continuous measurements
3. 'cholesterol', 'gluc', 'smoke', 'alco', 'active', 'cardio'] are binary or discrete measurements

Data Summary

1. Age group in days = approx 29 to 65 years. So all are adults
2. ap_hi has Negative & extreme values => min = -150, max = 16020. Impossible blood pressure. Appears as data entry issues
3. ap_lo also has Negative & extreme values => min = -70, max = 11000. Impossible blood pressure. Appears as data entry issues
4. height seems to have unrealistic value like min = 55 cms (too short for adult)
5. weight seems to have unrealistic value like min = 10 kg (too light for adult)
6. Needs cleaning.

2. Data Cleaning

Identify and remove invalid/illogical physiological values, then convert units and data types for interpretability and correctness.

2.1 Identify and handle redundant or invalid/illogical physiological values

2.1.1 Check the statistical summary of the data

The statistical summary (`df.describe()`) exposed clear data-entry errors:

- - `ap_hi` (systolic): min -150, max 16,020 — impossible blood pressure
- - `ap_lo` (diastolic): min -70, max 11,000 — impossible blood pressure
- - `height`: min 55 cm — too short for an adult
- - `weight`: min 10 kg — too light for an adult

2.1.2 Handle rows with invalid/illogical values

Cleaning rules applied (physiological bounds, per the assignment hint that BP <30 or >300 is rarely observed):

- - `ap_hi` between 30 and 300
- - `ap_lo` between 30 and 300
- - `ap_lo` < `ap_hi` (diastolic must be below systolic)
- - `height` between 100 and 250 cm
- - `weight` between 20 and 200 kg

Result: 1,349 rows removed → dataset reduced from 70,000 to 68,651 rows (~1.9% dropped). No missing values were present, so no imputation was needed.

2.1.3 Modify the representation of patient age and height

- - `age`: days → years (range now ≈ 29.6–65 years, confirming an all-adult cohort)
- - `height`: cm → metres (needed for BMI in Section 5)

2.2 Fix DataTypes

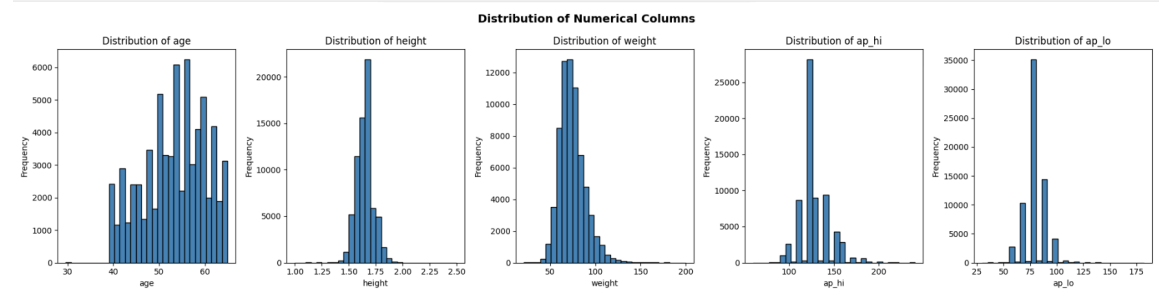
2.2.1 Review and fix the data types of all columns

- - Although columns like ``gender``, ``cholesterol``, ``gluc``, ``smoke``, ``alco``, ``active`` and ``cardio`` are stored as `int64`, they are semantically categorical
- - Their values represent discrete labels or ordinal levels, not continuous quantities.
- - For example, cholesterol values of 0, 1, 2 represent "normal", "above normal" and "well above normal". The numeric encoding is arbitrary and arithmetic operations on them are meaningless.
- - Converting these to category dtype:
 - - Prevents misleading mathematical operations (e.g. computing a mean of gender)
 - - More accurately represents the nature of the data
 - - Improves memory efficiency
 - - Makes the intent of the data explicit for anyone reading the code

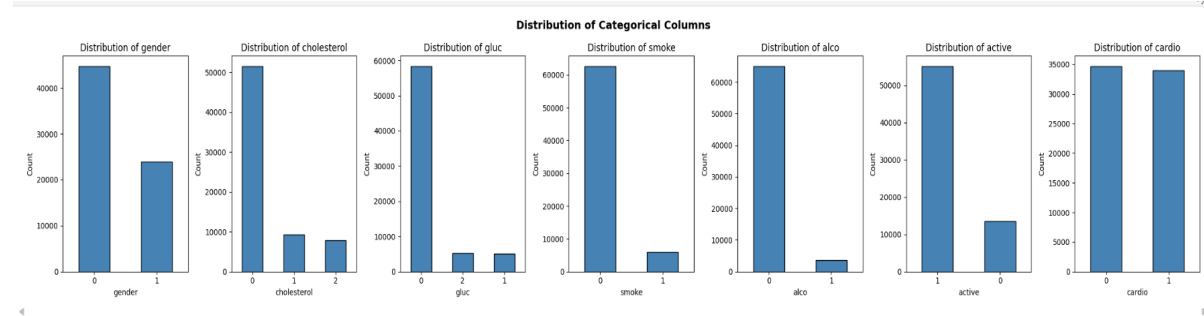
3. Exploratory Data Analysis

3.1 Perform univariate analysis

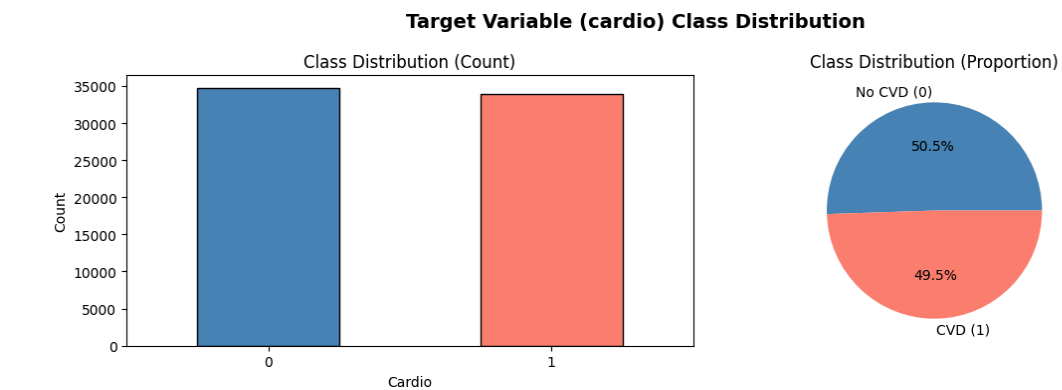
3.1.1 Visualise the numerical features



3.1.2 Visualise the categorical features

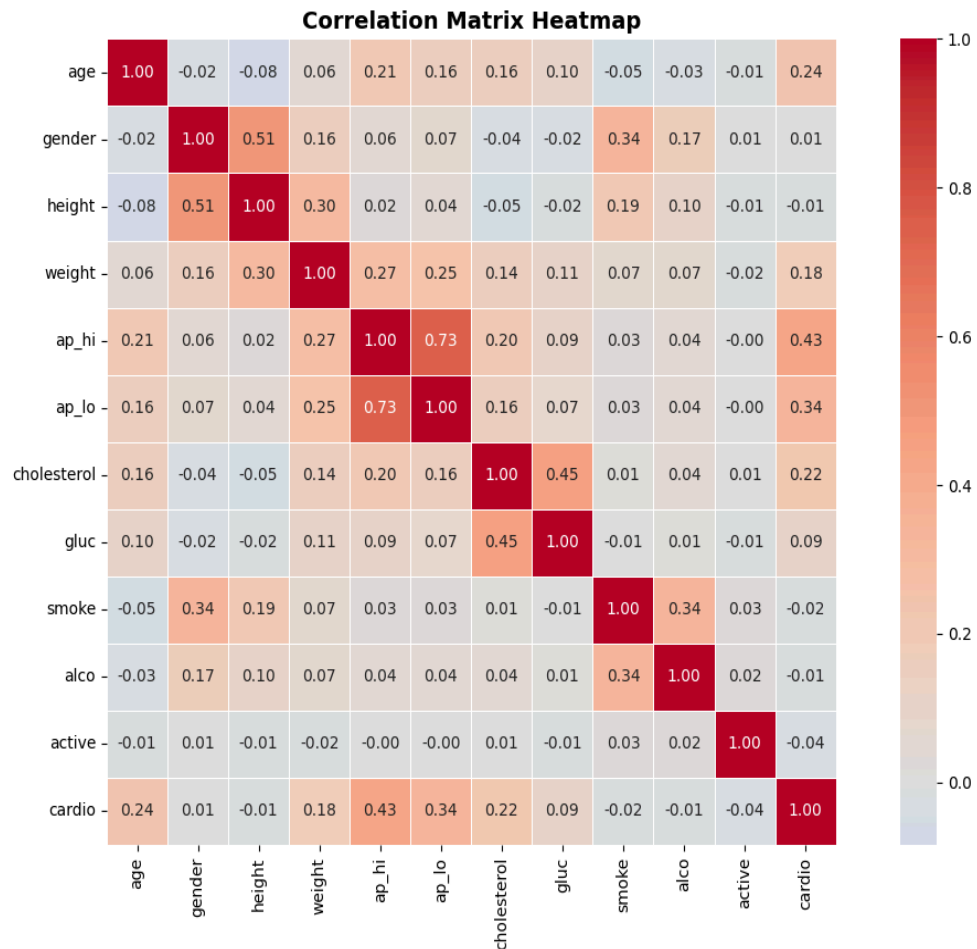


3.1.3 Check class distribution of the target feature



3.2 Perform correlation analysis

3.2.1 Visualise the correlation among numerical features



Correlation Matrix Heatmap — Interpretation

Strong Feature Correlations

- - `ap\_hi` to `ap\_lo` (0.73):** Strongest pair - systolic and diastolic blood pressure
- move together as expected physiologically.
- - `gender` to `height` (0.51):** Males tend to be taller - biological correlation.
- - `cholesterol` ↔ `gluc` (0.45):** High cholesterol often co-occurs with high glucose— common metabolic risk factors.
- - `smoke` to `alco` (0.34): Smokers tend to drink more - lifestyle behaviour correlation.

Correlations with Target Variable (**cardio**)

Feature	Correlation	Interpretation
ap_hi	0.43	Strongest predictor - high systolic BP strongly linked to CVD risk
ap_lo	0.34	Diastolic BP also significant
age	0.24	Older patients at higher CVD risk
cholesterol	0.22	High cholesterol linked to CVD
weight	0.18	Higher weight associated with CVD risk
gluc	0.09	Weak relationship with CVD
active	-0.04	Slight negative - physically active patients less at risk
smoke	-0.02	Surprisingly weak - possibly due to underreporting

Key Takeaways

1. **Blood pressure (**ap_hi**, **ap_lo**) dominates** as the strongest CVD

✓ EDA Summary

Dataset Overview

- 70,000 patients, 12 columns (after cleaning), balanced target: 50.5% No CVD / 49.5% CVD
- No missing values across any column

Numerical Features (Image 1)

Feature	Distribution	Key Observation
age	Multimodal, 30–65 years	Peaks at 50–60 — middle-aged population
height	Bell curve, 1.50–1.75m	Normal, realistic adult range
weight	Right skewed, peak 65–75kg	Some heavier outliers remaining
ap_hi	Discrete spikes at 120,130,140	Clinical rounding pattern
ap_lo	Discrete spikes at 70,80,90	Same clinical rounding

Categorical Features (Image 2)

Feature	Distribution	Key Observation
gender	65% female, 35% male	Slight gender imbalance
cholesterol	75% normal, 25% elevated	Majority healthy — minority high risk
gluc	85% normal, 15% elevated	Heavily skewed to normal
smoke	91% non-smokers	Very few smokers
alco	94% non-drinkers	Very few drinkers
active	80% physically active	Majority active

Target Variable (Image 3)

- Near perfectly balanced — **50.5% No CVD vs 49.5% CVD**
- No class imbalance treatment needed
- Accuracy is a valid evaluation metric

Correlation Analysis (Image 4)

Correlation Analysis (Image 4)

Strong feature correlations:

- `ap_hi` ↔ `ap_lo`: 0.73 — systolic/diastolic move together
- `gender` ↔ `height`: 0.51 — males tend to be taller
- `cholesterol` ↔ `gluc`: 0.45 — metabolic risk factors co-occur
- `smoke` ↔ `alco`: 0.34 — lifestyle behaviour correlation

Correlations with target (`cardio`):

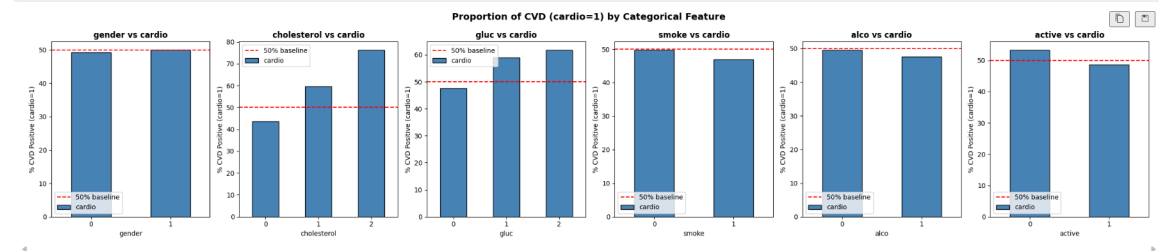
- `ap_hi` (0.43) and `ap_lo` (0.34) — strongest predictors
- `age` (0.24) — older patients at higher risk
- `cholesterol` (0.22) — elevated cholesterol linked to CVD
- `active` (-0.04) — physical activity is mildly protective
- `smoke`, `alco` (~0.00) — surprisingly uninformative

Key Takeaways

1. **Blood pressure is the dominant CVD predictor** — both systolic and diastolic
2. **Dataset is clean and balanced** — ideal for binary classification
3. **Lifestyle features (`smoke`, `alco`) are weak predictors** — likely due to underreporting bias in self-reported data
4. **BMI was engineered** from weight and height — weight and height subsequently dropped due to high multicollinearity (BMI ↔ weight: 0.85)
5. **Cholesterol and gluc** were recoded from 3 levels to binary (normal vs elevated) due to similar behaviour across levels 1 and 2

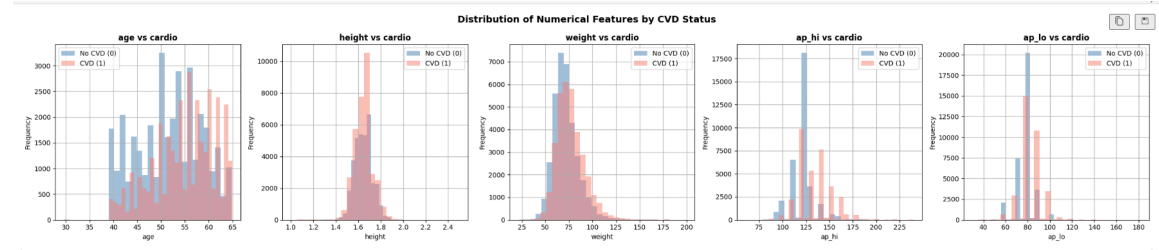
3.3 Perform bivariate analysis

3.3.1 Analyse categorical features



- `cholesterol` is the most informative categorical feature — CVD likelihood increases clearly from 45% (normal) → 58% (above) → 75% (well above normal)
- `gluc` shows a moderate effect — elevated glucose is associated with higher CVD risk but the signal is weaker than cholesterol
- `active` shows a small protective effect — inactive patients (0) have slightly higher CVD rates than active patients (1)
- `gender` shows virtually no difference between categories — not a useful predictor of CVD in this dataset
- `smoke` and `alco` are surprisingly uninformative — both hover near the 50% baseline, consistent with the near-zero correlations seen in the heatmap

3.3.2 Explore the relationships between numerical features and the target variable



- **ap_hi** and **ap_lo**: Show the clearest class separation — CVD patients have consistently higher blood pressure. Strongest numerical predictors.
- **age**: CVD patients skew older (50–65) vs non-CVD patients (40–55). Moderate predictive power.
- **weight**: Distributions largely overlap with slight rightward shift for CVD patients. Weak predictor.
- **height**: Almost identical distributions for both classes. Not a useful predictor of CVD risk.

4. Train-Test Split

4.1 Data Splitting

Objective: Separate features from the target and split the data into training and test sets for unbiased model evaluation.

4.1.1 Define feature and target variables

- X (features): all 11 predictor columns — age, gender, height, weight, ap_hi, ap_lo, cholesterol, gluc, smoke, alco, active
- y (target): cardio (presence of cardiovascular disease)
- Shape: X = (68,651 × 11), y = (68,651)

4.1.2 Split the data into train and test sets

- - Split 70% train / 30% test using train_test_split
- - random_state = 42 for reproducibility
- - stratify = y to preserve the class balance across both sets
- - Resulting sizes:
 - - Train: 48,055 rows
 - - Test: 20,596 rows
- - Indexes reset on all four sets (X_train, X_test, y_train, y_test) for clean downstream processing
- Class balance preserved
 - - Train: ~50.5% No-CVD / ~49.5% CVD
 - - Test: ~50.5% No-CVD / ~49.5% CVD
 - - Stratification kept the near-50/50 distribution identical in both sets, ensuring the test set is representative and metrics are not skewed by class imbalance.

5. Feature Engineering

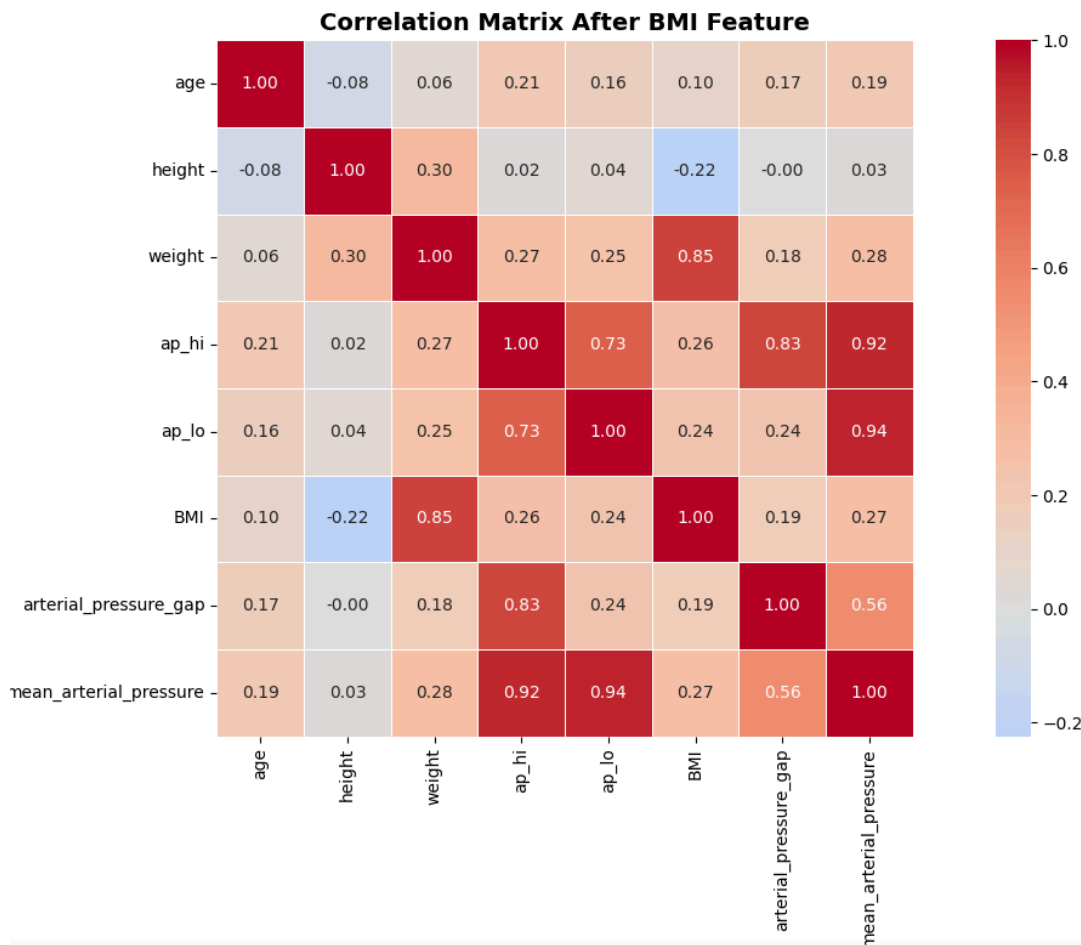
Objective: Derive clinically meaningful features, remove redundancy/multicollinearity, encode categoricals, and scale numerics, applying every transform consistently to train and test.

5.1 Create a new feature

5.1.1 Create a new feature BMI (Body Mass Index)

5.1.2 Perform correlation analysis

- - Created BMI = weight / height² — a standard, clinically validated CVD risk metric.
- - Also engineered two blood-pressure features for evaluation:
 - - arterial_pressure_gap = ap_hi – ap_lo (pulse pressure; >40 mmHg is unhealthy)
 - - mean_arterial_pressure = (ap_hi + 2×ap_lo) / 3 (clinical MAP formula)
- - Correlation check revealed strong multicollinearity, so redundant columns were dropped:
 - - weight — 0.85 correlation with BMI (BMI already encodes it)
 - - height — information already captured in BMI's height² term
 - - mean_arterial_pressure — 0.94 with ap_lo, 0.92 with ap_hi (no new information)
 - - arterial_pressure_gap — 0.83 with ap_hi; a linear combination the model can derive itself (unlike BMI's non-linear term)
- Note on leakage avoidance: BMI was computed and the drops applied separately on X_train and X_test (after the split), not on the combined data. Final feature set: 10 columns including BMI.



5.2 Combine Values in Categorical Columns

5.2.1 Combine Low-Frequency Categories

- - cholesterol (0/1/2 = normal / above / well-above): sparse levels 1 & 2 both signal elevated risk → merged into a single "1" (now ~75% / ~25%).
- - gluc: same logic → merged to binary (~85% / ~15%).
- - smoke (8.8%) and alco (5.3%) are sparse but already binary — no combining possible; other columns left unchanged.

5.3 Dummy variable creation

5.3.1 Create dummy variables for categorical columns

- - All categorical predictors (gender, cholesterol, gluc, smoke, alco, active) are now binary.
- - Applied `pd.get_dummies(drop_first=True)` for sklearn compatibility and to avoid the dummy-variable trap.
- - Encoded train and test separately, then asserted identical columns (alignment check passed). Result: 10 features each.

5.4 Feature scaling

5.4.1 Scale numerical features

- - Scaled only the genuine continuous features — age, ap_hi, ap_lo, BMI — using `MinMaxScaler` (binary dummies left as 0/1).
- - Scaler fit on training data only, then applied (transform) to test — preventing data leakage from the test distribution.
- - Post-scaling, all four features lie in $[0, 1]$.

6. Model Building

6.1 SVM Classifier

6.1.1 Define a Linear SVM classifier and fit it on the train set

- Trained an SVC with linear kernel, $C = 1.0$, `probability=True` (enables predict_proba via Platt Scaling), `random_state=42`.
- Training took ~1,232 s (20.5 min) — slow due to Platt Scaling running 5-fold CV internally.
- Model retained 31,420 support vectors (65.4% of training data), indicating a hard-to-separate, overlapping dataset.

6.1.2 Get the probability estimates on test set and predict class using a threshold

- - Used `predict_proba()` on the test set (~30 s) to get positive-class (CVD) probabilities.
- - Applied the default 0.5 cutoff → predicted 56.5% No-CVD / 43.5% CVD.
- - Noted SVM's $O(n^2)$ – $O(n^3)$ complexity and the Platt-scaling overhead make probability estimation expensive at this data size.

6.1.3 Predict the class labels using the `predict()` function

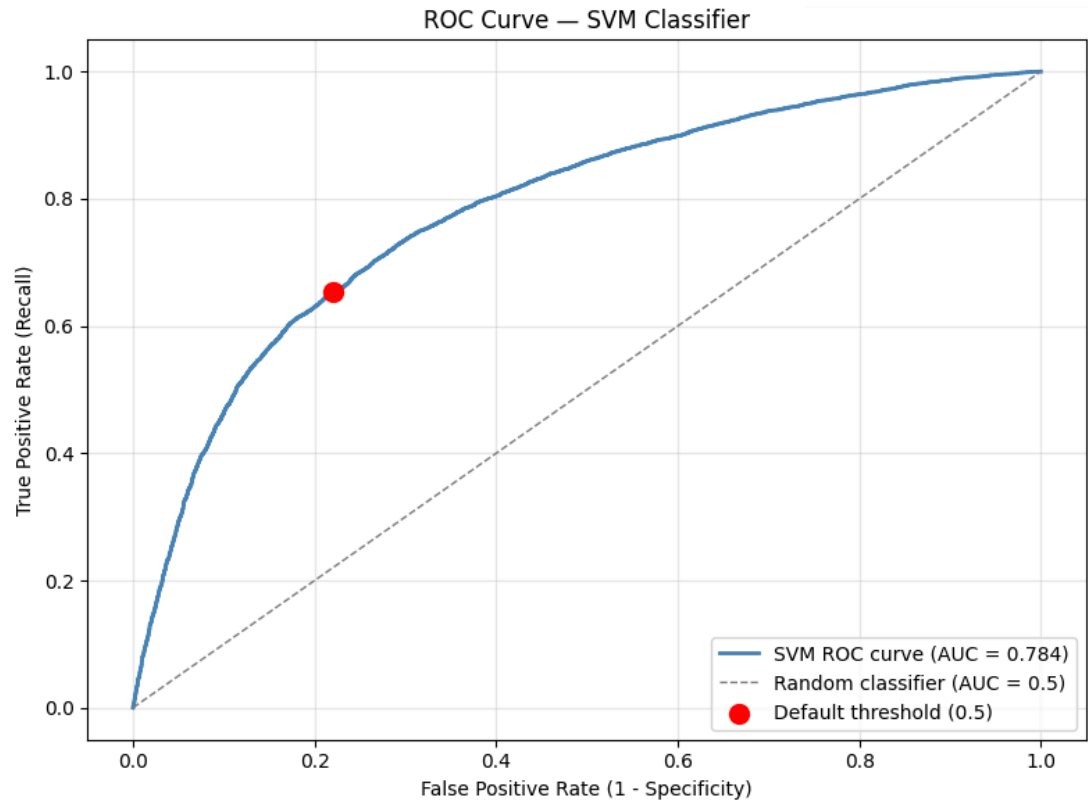
- - `predict()` (native SVM decision boundary at score 0.0) gave 59.2% No-CVD / 40.8% CVD.
- - Slightly more conservative than `predict_proba ≥ 0.5`, because a 0.5 probability does not map exactly to a 0.0 decision score under Platt's non-linear calibration.

6.1.4 Calculate performance metrics for both the above methods

- - `predict_proba ≥ 0.5`: Accuracy 71.7%, Precision 74.3%, Recall 65.4%, F1 0.70.
- - `predict()`: Accuracy 71.5%, Precision 75.7%, Recall 62.5%, F1 0.68.
- - Both underpredict CVD vs the true ~49.5%.
- - Chose `predict_proba ≥ 0.5` — higher recall (294 more CVD caught) plus the flexibility to tune the threshold.
- - Clinical framing: False Negatives (3,530) are fatal vs False Positives (2,299) merely an irritant → recall is the priority metric.

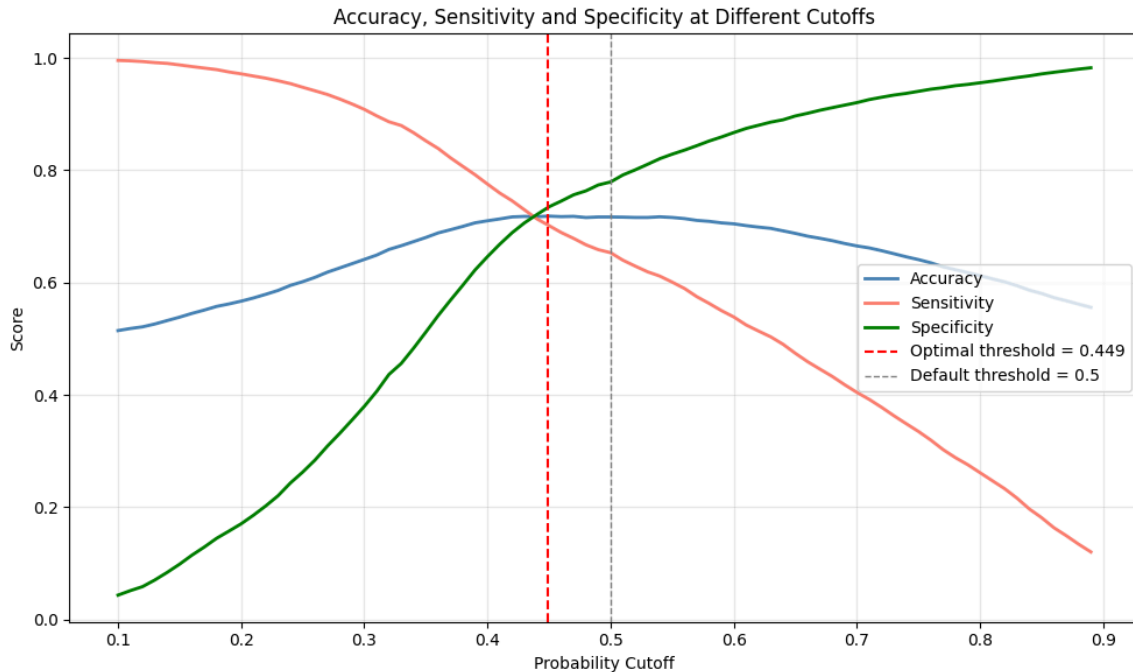
6.1.5 Plot the ROC curve

- - AUC = 0.78 — clearly above the 0.5 random baseline; good discriminative ability.
- - Youden's J identified an optimal threshold ≈ 0.449 (TPR 0.703, FPR 0.266), well below 0.5 — consistent with prioritising recall.



6.1.6 Plot for accuracy, sensitivity, specificity at different probability cutoffs

- - Sensitivity and specificity trade off inversely; accuracy plateaus ~71–72% across 0.44–0.50.
- - Sensitivity–specificity crossover ~0.44 (both ~0.72), matching the Youden optimum and confirming 0.449 as a balanced choice.
- - At 0.449: sensitivity ~0.70, specificity ~0.73 — versus default 0.5 (sensitivity ~0.65, specificity ~0.78).

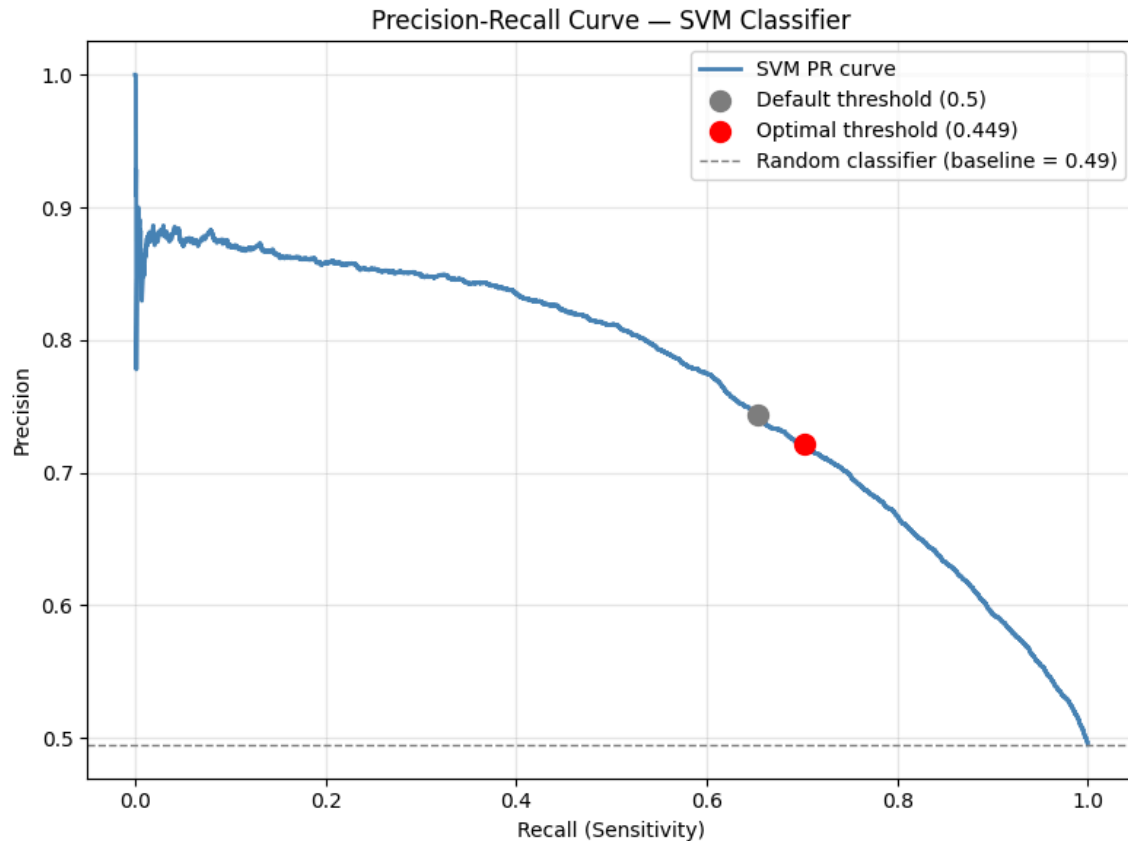


6.1.7 Assign classes based on the optimal cutoff and evaluate

- - Test: Accuracy 72%, Recall (CVD) 70%, Precision 72%, F1 0.71.
- - Train: Accuracy 73%, Recall 72%, Precision 73%, F1 0.72.
- - Train-test gap within ~2% → no meaningful overfitting; model generalises well.
- - Recall improved over default 0.5 (~65% → ~70%), accepting more false positives — the correct clinical trade-off.

6.1.8 Plot precision-recall curve

- - PR curve chosen as more clinically meaningful than ROC (focuses on the positive CVD class).
- - Curve stays well above the 0.49 baseline across all thresholds → real predictive value.
- - Moving from 0.5 → 0.449 shifts the operating point rightward: recall 0.65 → 0.70 (+5%), precision 0.74 → 0.72 (-2%) — accepted to minimise missed CVD cases.



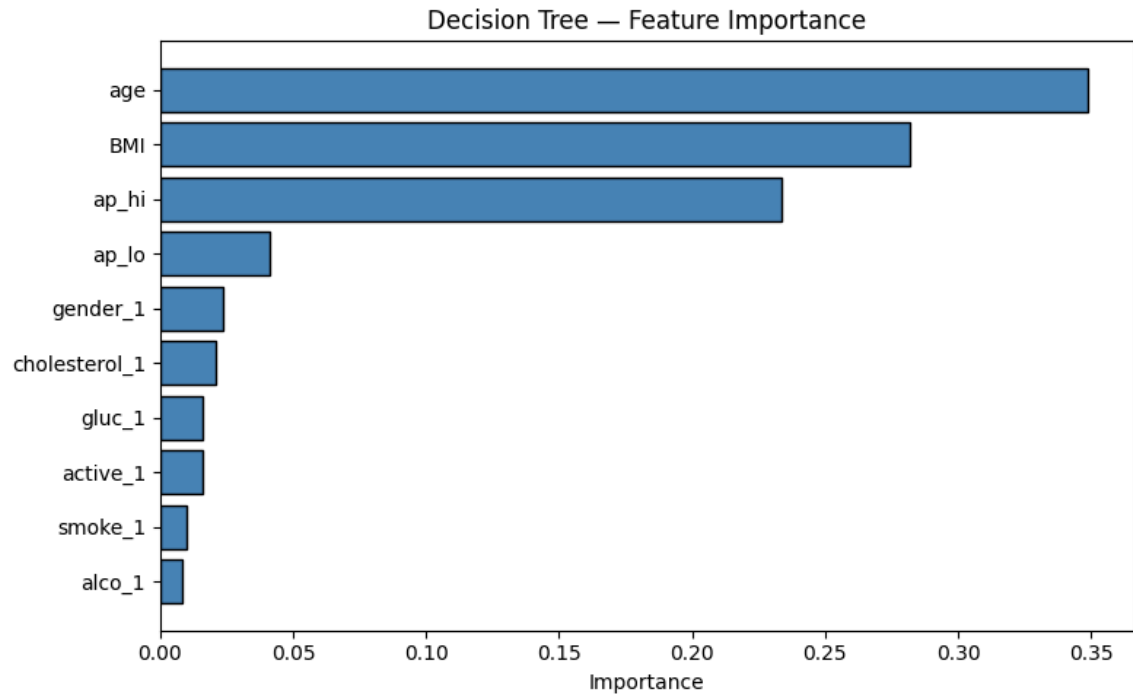
6.2 Decision Tree Classifier

6.2.1 Define a Decision Tree classifier and fit it on the train set

- - Trained an unconstrained DecisionTreeClassifier (random_state=42) on the 10 features in <1 s — far faster than the SVM's 20 minutes.
- - Resulting tree was depth 66 with 13,387 leaves — a clear early sign of an over-grown, memorising model.

6.2.2 Get feature importance scores

- - Top drivers: age (0.35), BMI (0.28), ap_hi/systolic BP (0.23) — together ~86% of total importance.
- - ap_lo (0.04) and all categorical dummies (gender, cholesterol, gluc, etc.) contributed marginally (<0.05 each).
- - Consistent with the EDA/correlation findings that BP, age and body mass dominate CVD risk.



6.2.3 Predict the class probabilities on the test set

- - `predict_proba()` ran in ~0.1 s.
- - Probabilities are almost all 0.0 or 1.0 — the near-pure leaves leave virtually no graded confidence (a red flag).

6.2.4 Make prediction based on default cutoff value of 0.5 on testing data

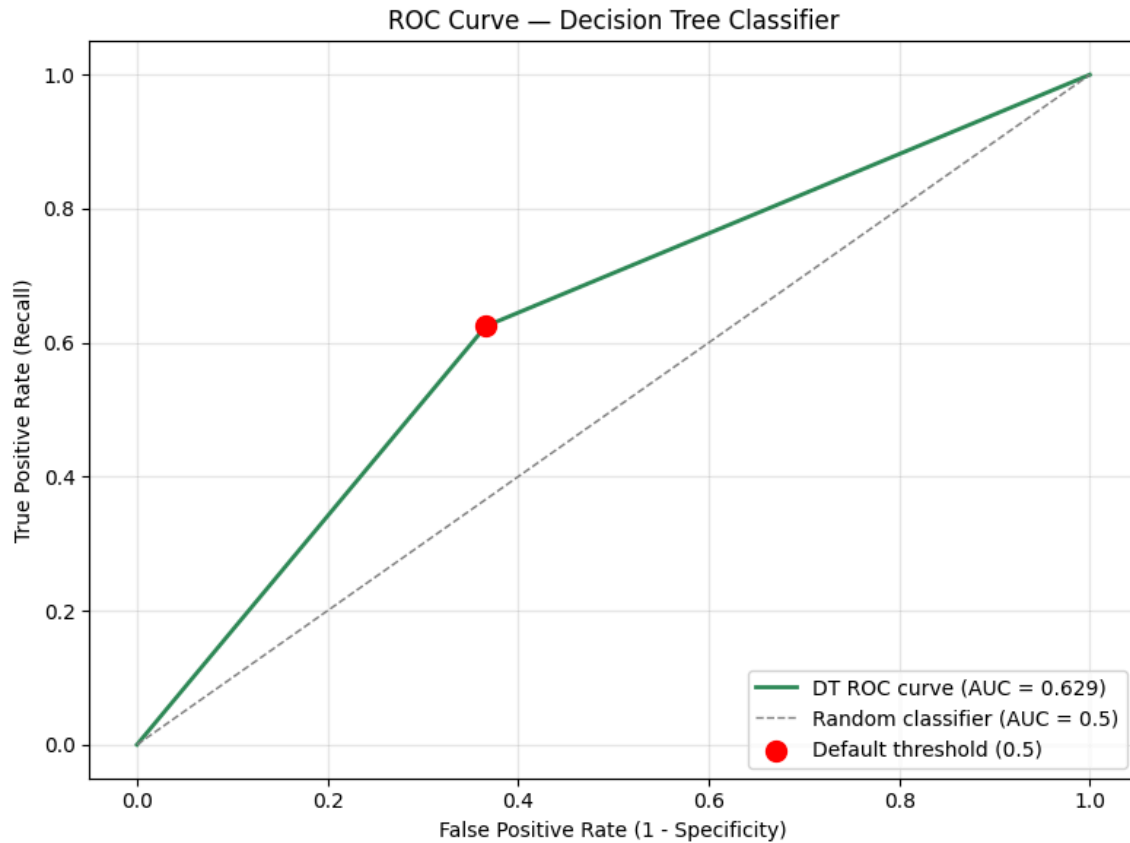
- Predicted 50.6% No-CVD / 49.4% CVD — very close to the true class balance.

6.2.5 Evaluate the performance of the model

- - Train: 100% accuracy / recall / precision (confusion matrix essentially perfect).
- - Test: ~63% accuracy, recall 62%, precision 63%.
- - Train-test gap ~37–38% → textbook severe overfitting: the tree memorised all 48,055 training patients but generalises poorly.

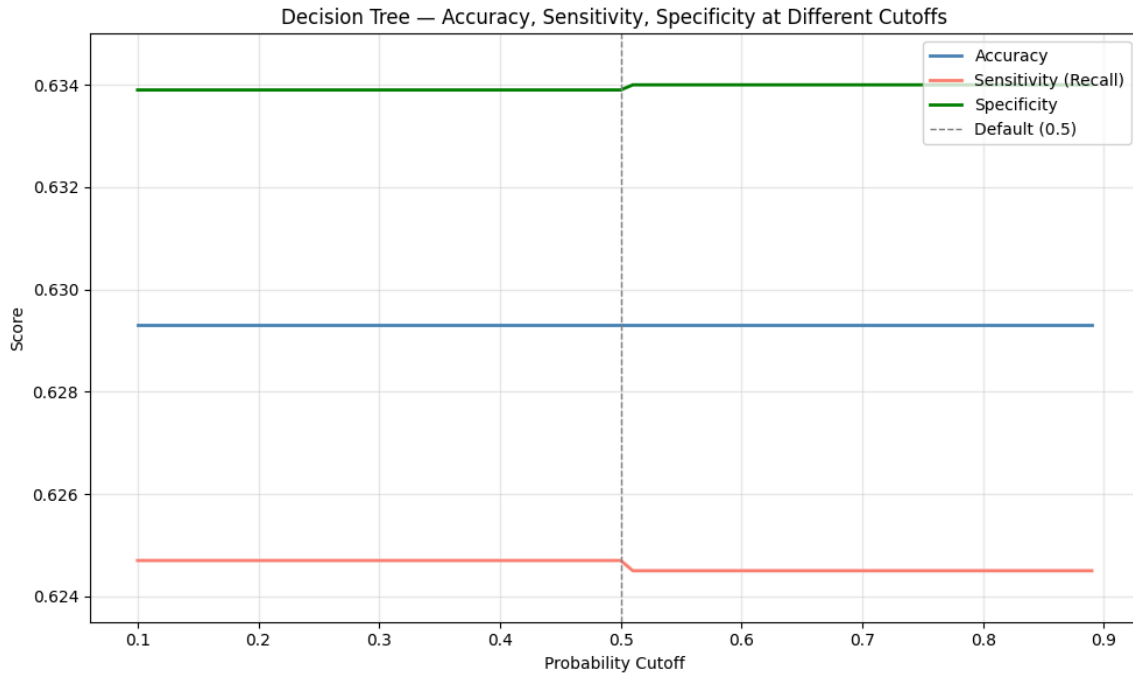
6.2.6 Plot the ROC curve

- - AUC = 0.63 — barely above random (0.5) and far below SVM's 0.78.
- - Curve is angular/2-segment (only a few distinct probability values), the visual signature of overfitting.



6.2.7 Plot for accuracy, sensitivity, specificity at different probability cutoffs

- - Curves stay nearly flat (~ 0.62 – 0.63) across all thresholds with a single step near 0.5.
- - Because probabilities are concentrated at 0/1, there is almost nothing to threshold over $\rightarrow$ cutoff tuning gives negligible benefit ($< 1\%$).

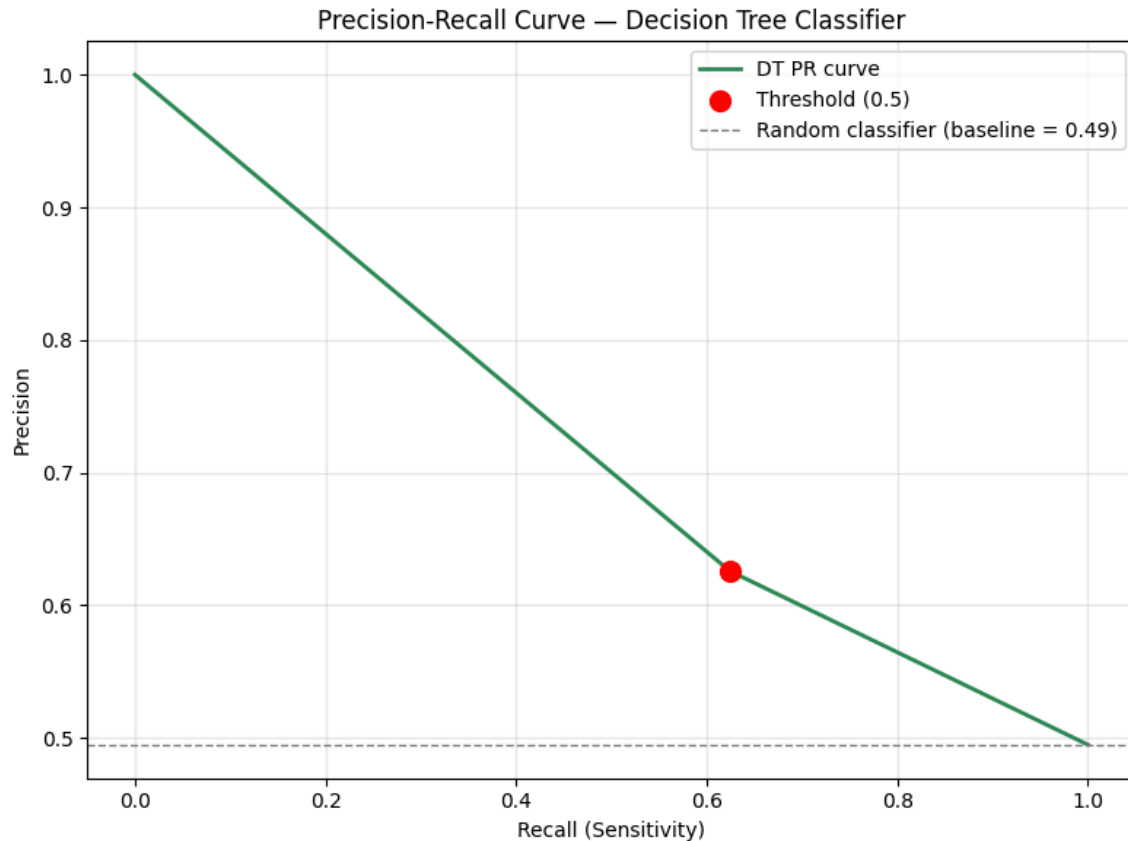


6.2.8 Assign classes based on the optimal cutoff and evaluate

- - Youden's optimal threshold = 0.5000 — identical to default.
- - Train vs test unchanged (100% vs 63%) → threshold tuning cannot fix overfitting; the problem is the tree's structure, not the cutoff. Motivates GridSearchCV in 6.3.

6.2.9 Plot precision-recall curve

- - Near-linear PR curve (straight line from (0,1) to (1, 0.49)) due to the binary-like probabilities.
- - At 0.5: precision $\approx$ recall $\approx$ 0.63 — well below SVM's 0.7–0.9 precision range; barely above the 0.49 baseline.



6.2.10 Build another model of your choice.

- - Built a 100-tree Random Forest (fit in ~3.6 s) to test whether ensemble averaging fixes the overfitting.
- - Test: accuracy 69%, recall 68%, AUC 0.75 vs DT's 63% / 62% / 0.63 — a meaningful improvement (+6 pts accuracy/recall, +0.12 AUC).
- - Still 100% train accuracy → variance reduced but not eliminated; a partial fix, reinforcing the need for hyperparameter tuning (6.3).

6.3 Hyperparameter Tuning

6.3.1 Use grid search to find the best hyperparameter values

- - Tuned 4 hyperparameters via GridSearchCV (5-fold CV, scoring = recall, the clinically priority metric):
 - - max_depth [3–20], min_samples_split [2–100], min_samples_leaf [1–50], max_features [sqrt, log2, None]
- - 450 combinations → 2,250 fits, completed in ~174 s (2.9 min).
- - Best params: max_depth=3, max_features='sqrt', min_samples_leaf=1, min_samples_split=2 — best CV recall 0.731.

6.3.2 Build a decision tree model based on hyperparameter tuning results

- - max_depth=3 was the dominant regularising parameter (others stayed permissive).
- - Tree shrank from depth 66 / 13,387 leaves → depth 3 / 8 leaves (~99.9% fewer leaves).
- - Each leaf now averages ~6,000 training samples (vs ~3.6) → moved from memorising patients to learning generalisable rules.

6.3.3 Using the tuned model, make predictions and evaluate

- - Train: accuracy 71%, recall 73%, precision 69%.
- - Test: accuracy 70%, recall 72%, precision 68%, AUC 0.73.
- - Train-test gap collapsed from ~37% → ~1% — healthy generalisation matching the SVM.
- - Notably, test performance also improved despite the far simpler tree: recall 62% → 72% (+10 pts), AUC 0.63 → 0.73. Confirms the original tree's complexity was pure noise-fitting.

6.3.4 Optionally, use grid search to find the best hyperparameter values for SVM

- - SVM tuning — not run (justified): a single SVC(probability=True) fit takes ~20 min; even a small 12-combination grid × 5-fold CV would run for days. RBF kernel discussed conceptually (could capture non-linear boundaries but is even slower on 48k rows).
- - Random Forest tuned instead (feasible — trains in seconds):
 - - Grid of n_estimators / max_depth / min_samples_leaf / max_features → 120 combos, 600 fits (~11 min).
 - - Best params: min_samples_leaf=10, max_depth=None, n_estimators=100 — regularises via larger leaf nodes rather than depth limits.
 - - Test: accuracy 73%, recall 68%, AUC 0.791 (train 77% / 73%) — gap ~4–5%, well controlled.
 - - Highest test AUC (0.791) of all models, edging out the SVM (0.784) and tuned DT (0.734).

7. Final Model Evaluation and Selection

7.1 Evaluate the final models

7.1.1 Make final predictions and evaluate

All three tuned candidates evaluated on the test set at their chosen cutoffs:

- - SVM (cutoff 0.449): Test Accuracy 71.9%, Recall 70.3%, Precision 72.1%, F1 0.712, AUC 0.784 — smallest train-test gap (~0.9%), but ~22 min to train.
- - Tuned Decision Tree (cutoff 0.5): Test Accuracy 69.7%, Recall 71.7% (highest), Precision 68.5%, F1 0.701, AUC 0.734 — fully interpretable (8 rules).

- - Tuned Random Forest (cutoff 0.5): Test Accuracy 72.5% (highest), Recall 68.0% (lowest), Precision 74.2%, F1 0.710, AUC 0.791 (highest).
- - Key trade-off: no model wins on every metric — DT maximises recall (catches most CVD), RF maximises accuracy/AUC (best ranking), SVM is the most stable but impractical to retrain.

7.2 Conclusion

7.2.1 Model Cards

- - Documented a model card for each candidate (overview, intended use, data/features, configuration, performance, limitations) per the supervised-ML template.
- - SVM card: linear kernel, C=1.0, Platt-calibrated, cutoff 0.449; batch use only, not real-time; weakness = very slow training, high support-vector fraction.
- - Decision Tree card: depth-3, 8-leaf transparent rules; highest recall; primary clinical recommendation.
- - Random Forest card: 100 trees, min_samples_leaf=10; best AUC/accuracy but lowest recall and least interpretable; small residual ~4% gap.

7.2.2 Conclusions and Outcomes

- - EDA insights: balanced data (50.5/49.5) → accuracy valid; blood pressure strongest CVD correlate (ap_hi 0.43, ap_lo 0.34), then age (0.24) and cholesterol (0.22); smoke/alco near-zero (likely self-report underreporting); active mildly protective.
- - Feature engineering: age→years, height→metres; BMI engineered then weight/height dropped (0.85 collinearity); cholesterol & gluc levels merged to binary; arterial_pressure_gap and mean_arterial_pressure tested but dropped (linear combination / 0.92–0.94 collinearity).
- - Overfitting & tuning: baseline DT severely overfit (100% train vs 63% test, 37% gap, depth 66 / 13,387 leaves); GridSearchCV (recall-scored) fixed it — max_depth=3 shrank the gap to ~1% and improved recall (62%→72%); RF tuned via min_samples_leaf=10, gap 31%→4%, AUC 0.753→0.791. SVM generalised well from the start (margin regularisation).
- - Data / linearity: 68,651 records were sufficient for stable training; a linear SVM was competitive (AUC 0.784) but not best — 65% support vectors signal class overlap; tree ensembles modestly outperformed, indicating mild non-linear structure.
- - Final choice: Tuned Decision Tree = primary model (highest recall + fully explainable 8-rule structure aligns with CardioCare's "don't miss CVD" priority); Tuned Random Forest = secondary for population-level risk ranking.
- - Business outcome: the pipeline flags ~70–72% of true CVD patients for priority consultation while clearing a similar share of low-risk patients — directly supporting the goal of optimising doctors' consultation time.

